

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

```
df = pd.read_csv("Customer churn.csv")
df.head()
```

	customerID	gender	SeniorCitizen	Partner	Dependents	tenure
0	7590-VHVEG	Female	0	Yes	No	1
1	5575-GNVDE	Male	0	No	No	34
2	3668-QPYBK	Male	0	No	No	2
3	7795-CF0CW	Male	0	No	No	45
4	9237-HQITU	Female	0	No	No	2

	MultipleLines	InternetService	OnlineSecurity	...
0	No phone service	DSL	No	...
1	No	DSL	Yes	...
2	No	DSL	Yes	...
3	No phone service	DSL	Yes	...
4	No	Fiber optic	No	...

	TechSupport	StreamingTV	StreamingMovies	Contract
0	No	No	No	Month-to-month
1	No	No	No	One year
2	No	No	No	Month-to-month
3	Yes	No	No	One year
4	No	No	No	Month-to-month

	PaymentMethod	MonthlyCharges	TotalCharges	Churn
0	Electronic check	29.85	29.85	No

1	Mailed check	56.95	1889.5	No
2	Mailed check	53.85	108.15	Yes
3	Bank transfer (automatic)	42.30	1840.75	No
4	Electronic check	70.70	151.65	Yes

[5 rows x 21 columns]

df.info()

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 7043 entries, 0 to 7042
Data columns (total 21 columns):
#   Column                Non-Null Count  Dtype
---  -
0   customerID            7043 non-null   object
1   gender                 7043 non-null   object
2   SeniorCitizen          7043 non-null   int64
3   Partner                7043 non-null   object
4   Dependents             7043 non-null   object
5   tenure                 7043 non-null   int64
6   PhoneService           7043 non-null   object
7   MultipleLines           7043 non-null   object
8   InternetService        7043 non-null   object
9   OnlineSecurity         7043 non-null   object
10  OnlineBackup           7043 non-null   object
11  DeviceProtection       7043 non-null   object
12  TechSupport            7043 non-null   object
13  StreamingTV            7043 non-null   object
14  StreamingMovies        7043 non-null   object
15  Contract               7043 non-null   object
16  PaperlessBilling       7043 non-null   object
17  PaymentMethod          7043 non-null   object
18  MonthlyCharges         7043 non-null   float64
19  TotalCharges           7043 non-null   object
20  Churn                  7043 non-null   object
dtypes: float64(1), int64(2), object(18)
memory usage: 1.1+ MB
```

#Replacing blanks with 0 as tenure is 0 and no total charges are recorded

```
df["TotalCharges"] = df["TotalCharges"].replace(" ", "0")
df["TotalCharges"] = df["TotalCharges"].astype("float")
```

df.info()

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 7043 entries, 0 to 7042
Data columns (total 21 columns):
#   Column                Non-Null Count  Dtype
---  -
```

```

0    customerID      7043 non-null object
1    gender          7043 non-null object
2    SeniorCitizen    7043 non-null int64
3    Partner          7043 non-null object
4    Dependents       7043 non-null object
5    tenure           7043 non-null int64
6    PhoneService     7043 non-null object
7    MultipleLines    7043 non-null object
8    InternetService  7043 non-null object
9    OnlineSecurity   7043 non-null object
10   OnlineBackup     7043 non-null object
11   DeviceProtection 7043 non-null object
12   TechSupport      7043 non-null object
13   StreamingTV      7043 non-null object
14   StreamingMovies  7043 non-null object
15   Contract         7043 non-null object
16   PaperlessBilling 7043 non-null object
17   PaymentMethod    7043 non-null object
18   MonthlyCharges   7043 non-null float64
19   TotalCharges     7043 non-null float64
20   Churn            7043 non-null object

```

dtypes: float64(2), int64(2), object(17)

memory usage: 1.1+ MB

```
df.isnull().sum().sum()
```

```
np.int64(0)
```

```
df.describe()
```

	SeniorCitizen	tenure	MonthlyCharges	TotalCharges
count	7043.000000	7043.000000	7043.000000	7043.000000
mean	0.162147	32.371149	64.761692	2279.734304
std	0.368612	24.559481	30.090047	2266.794470
min	0.000000	0.000000	18.250000	0.000000
25%	0.000000	9.000000	35.500000	398.550000
50%	0.000000	29.000000	70.350000	1394.550000
75%	0.000000	55.000000	89.850000	3786.600000
max	1.000000	72.000000	118.750000	8684.800000

```
df["customerID"].duplicated().sum()
```

```
np.int64(0)
```

#Converted 0 and 1 values of senior citizen to yes/no to make it easier to understand

```

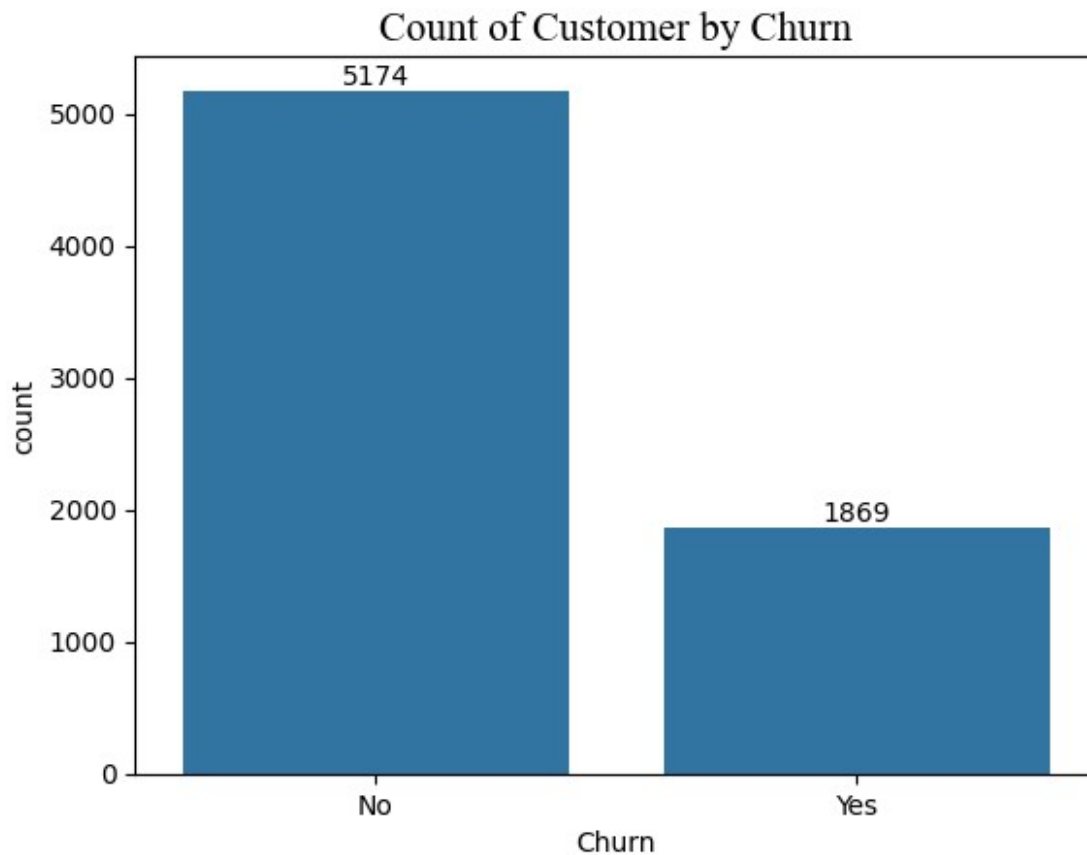
def conv(value):
    if value == 1:
        return "Yes"
    else:

```

```

        return "No"
df["SeniorCitizen"] = df["SeniorCitizen"].apply(conv)
ax = sns.countplot(x="Churn", data = df)
ax.bar_label(ax.containers[0])
plt.title("Count of Customer by Churn",fontdict={"fontname": "Times
New Roman", "fontsize": 15})
plt.show()

```

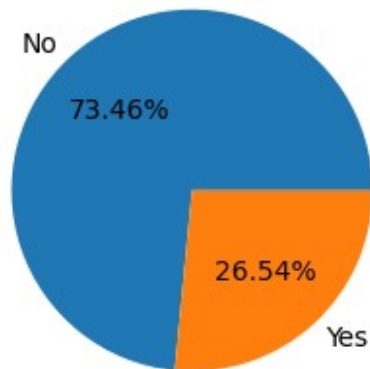


```

plt.figure(figsize=(3,4))
gb = df.groupby("Churn").agg({"Churn":"count"})
plt.pie(gb["Churn"],labels=gb.index,autopct="%1.2f%%")
plt.title("Percentage of Chured Customeres",fontdict={"fontname":
"Times New Roman", "fontsize": 15})
plt.show()

```

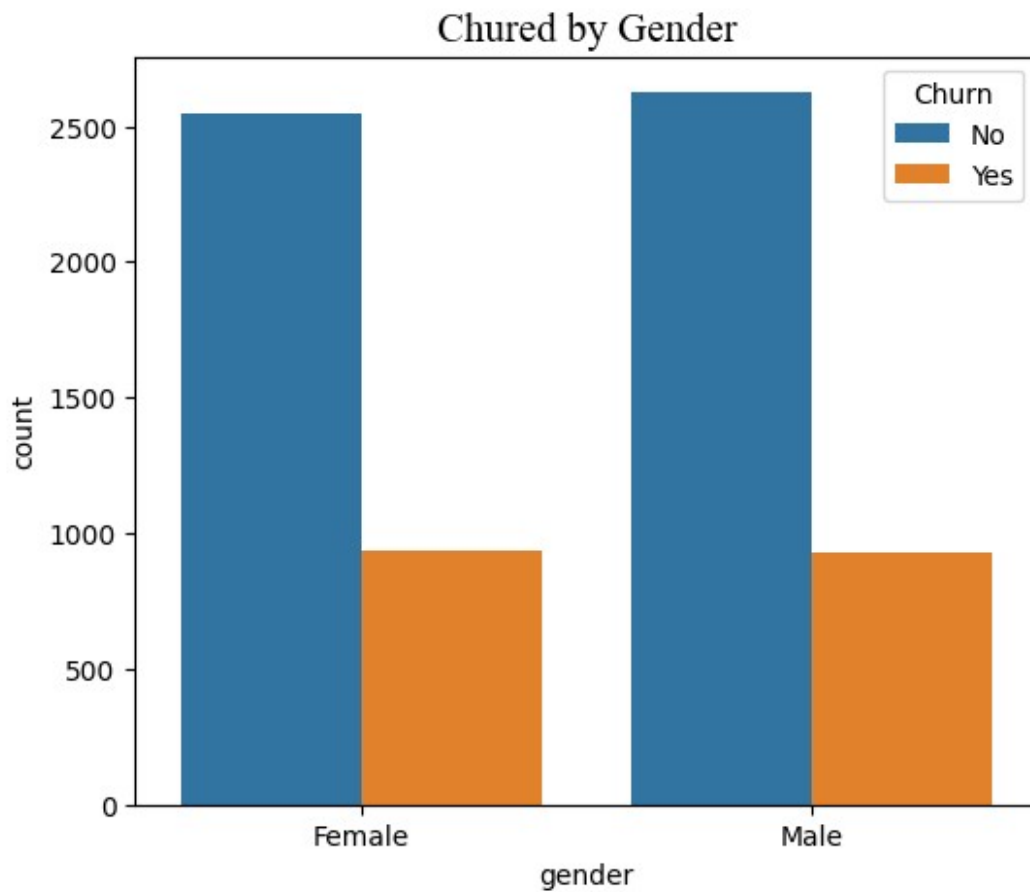
Percentage of Churned Customers



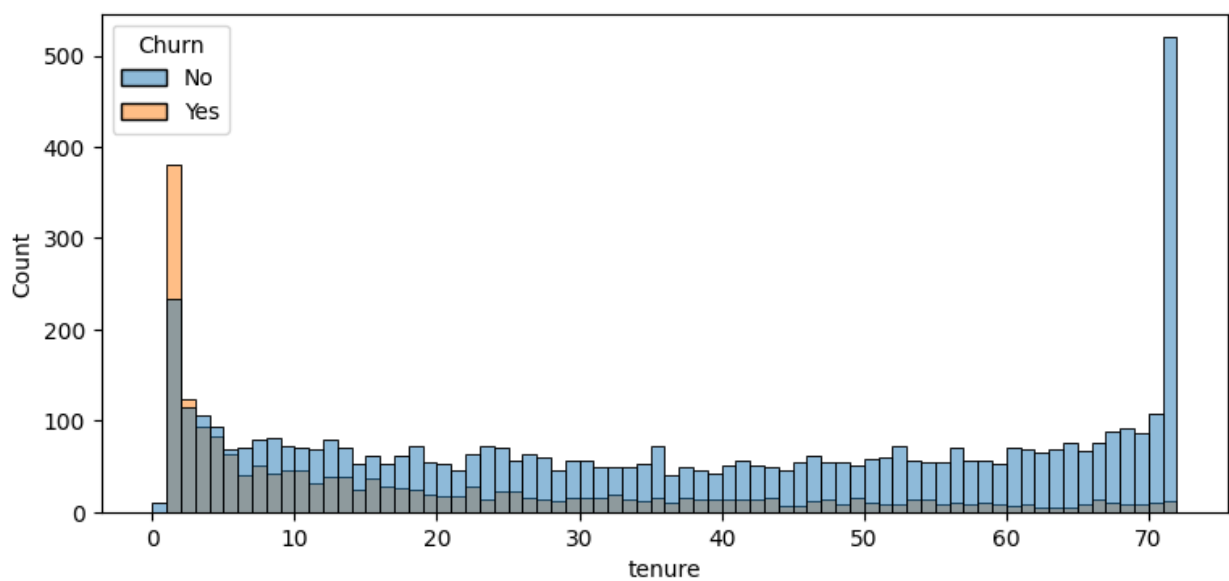
#From the given pie chart we can conclude that 26.54% of our customers have churned out.

#Not let's explore the reason behind it

```
plt.figure(figsize=(6,5))
sns.countplot(x = "gender",data = df, hue = "Churn")
plt.title("Churned by Gender",fontdict={"fontname": "Times New Roman",
"fontsize": 15})
plt.show()
```

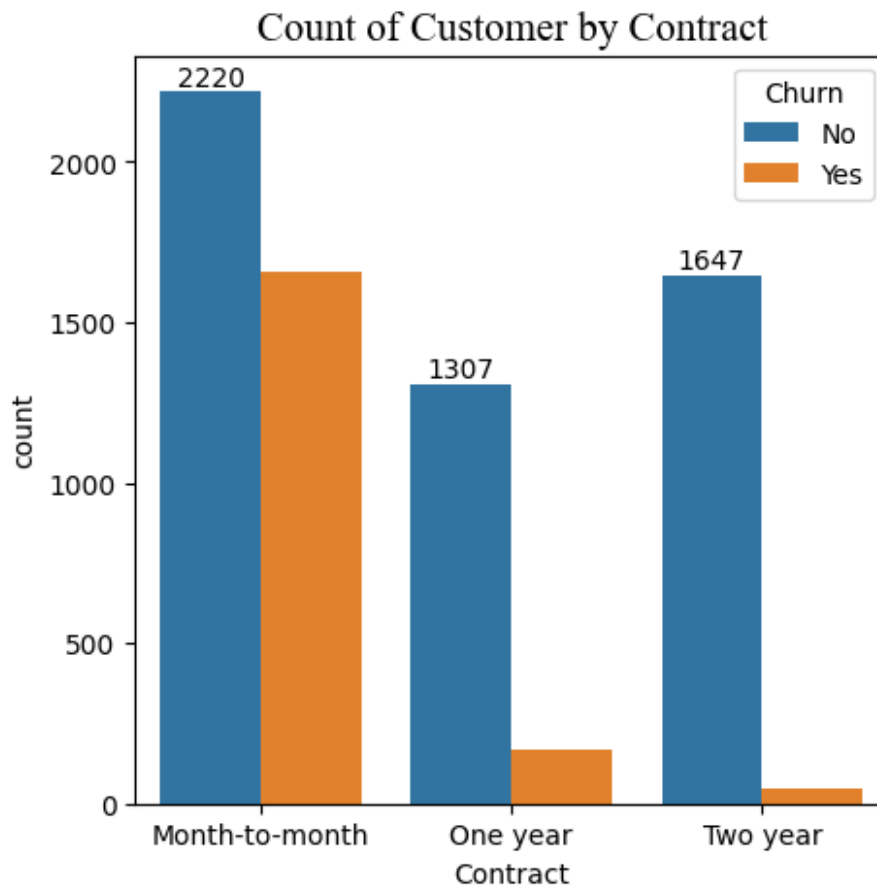


```
plt.figure(figsize=(9,4))  
sns.histplot(x="tenure", data=df, bins=72, hue="Churn")  
plt.show()
```



#People who have used our services for a long time have stayed and people who have used our services #1 or 2 Months have churned

```
plt.figure(figsize=(5,5))
ax = sns.countplot(x = "Contract", data=df, hue="Churn")
ax.bar_label(ax.containers[0])
plt.title("Count of Customer by Contract", fontdict={"fontname": "Times New Roman", "fontsize": 15})
plt.show()
```



#People who have month to month contract are likely to churn then form those who have 1 or 2 years or contract

```
# Columns you specified
cols = [
    'PhoneService', 'MultipleLines', 'InternetService',
    'OnlineSecurity', 'OnlineBackup', 'DeviceProtection',
    'TechSupport', 'StreamingTV', 'StreamingMovies'
]

# Subplot grid setup
```

```

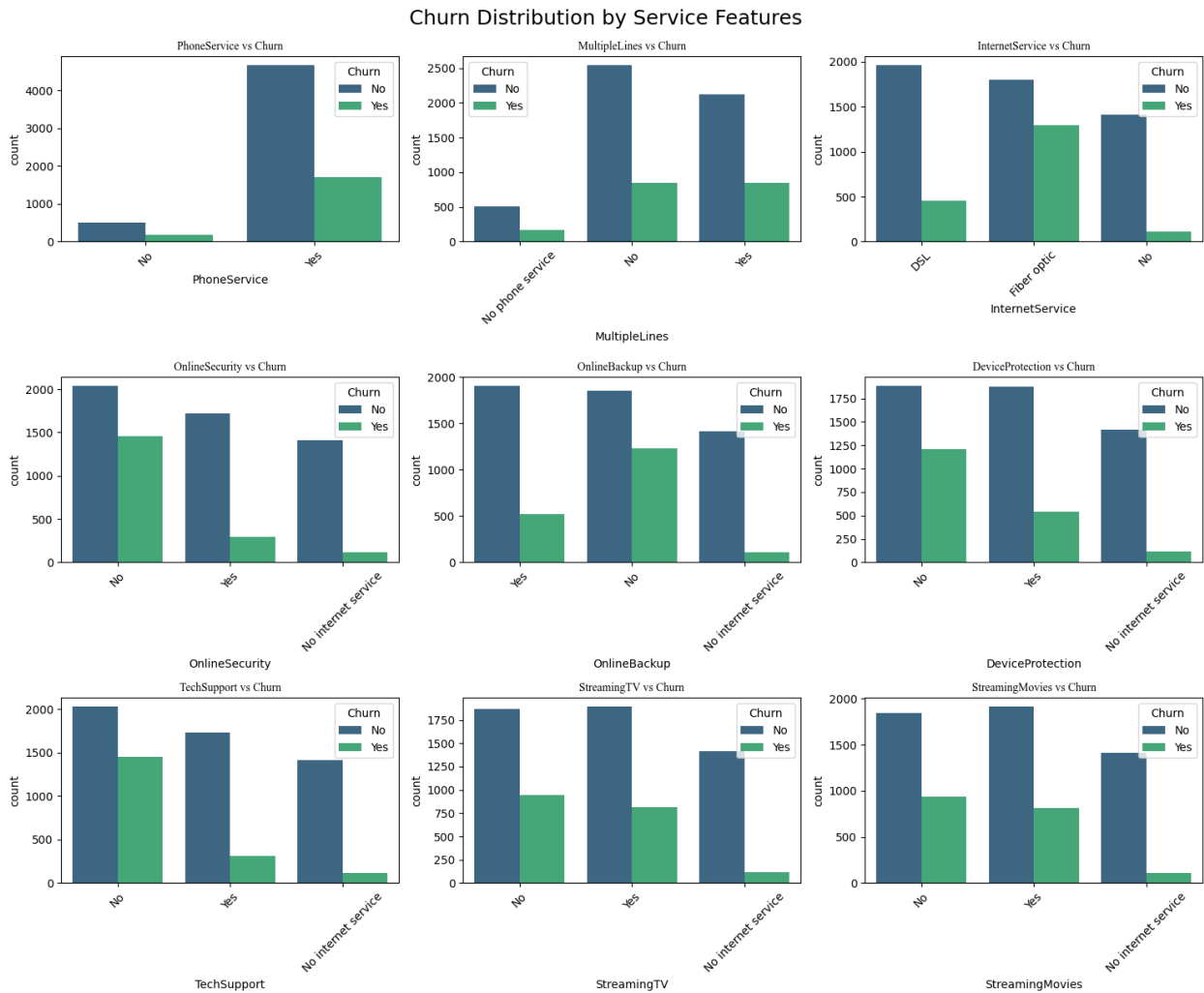
n_cols = 3
n_rows = -(-len(cols) // n_cols) # Ceiling division

# Create subplots
plt.figure(figsize=(5 * n_cols, 4 * n_rows))

for i, col in enumerate(cols, 1):
    plt.subplot(n_rows, n_cols, i)
    sns.countplot(data=df, x=col, hue="Churn", palette="viridis")
    plt.title(f"{col} vs Churn", fontdict={"fontname": "Times New
Roman", "fontsize": 10})
    plt.xticks(rotation=45)
    plt.tight_layout()

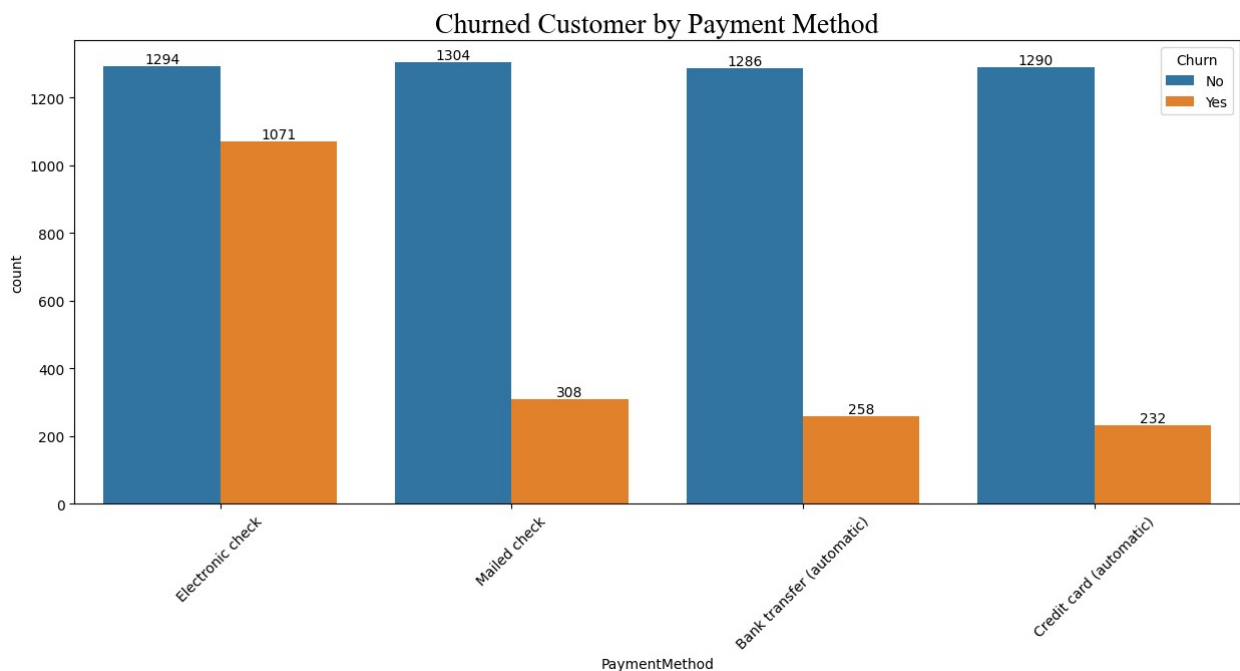
plt.suptitle("Churn Distribution by Service Features", fontsize=18,
y=1.02)
plt.show()

```



#The churn rate is highest for Fiber optic internet and without internet service, with tech support and online security significantly reducing churn. Multiple lines and streaming services show a moderate churn impact. Offering value-added services like Device Protection and Online Backup also correlates with lower churn.

```
plt.figure(figsize=(15,6))
ax = sns.countplot(x = "PaymentMethod", data=df, hue="Churn")
ax.bar_label(ax.containers[0])
ax.bar_label(ax.containers[1])
plt.title("Churned Customer by Payment Method", fontdict={"fontname":
"Times New Roman", "fontsize": 20})
plt.xticks(rotation = 45)
plt.show()
```



#Customer is likely to churn when he is using electronic check as a payment method